

Data Structures and Algorithms for Problem Solving (CS1.304)

Slip Test (10 Aug 2026)

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Puzzle 1 [3 marks]

Suppose that you have coins of four different denominations:

1, 6, 8, 13

There are unlimited coins of denomination 1, 6, 8, but only **two** coins of denomination 13.

- Obtain an amount **45** with the minimum number of coins.
- Obtain an amount **59** with the minimum number of coins.
- Obtain an amount **1000** with the minimum number of coins.

Puzzle 2 [7 marks]

You are given n coin denominations $c_1, c_2, c_3, \dots, c_n$, an amount A , and a number k (all numbers are positive integers) such that:

- $c_1 = 1$.
- $c_1 < c_2 < c_3 < \dots < c_n$.
- There are unlimited coins of denomination $c_1, c_2, c_3, \dots, c_{n-1}$ but only k coins of the denomination c_n .

Devise an algorithm to find the minimum total coins needed to add up to A . You shall be graded on your code / pseudocode, its explanation / analysis, and its runtime (the quicker, the better).

Thank You!

